

August 10, 2026

Editor-in-Chief
Empirical Software Engineering

Dear Editor,

Subject: Submission of manuscript
“Recovery-Channel Prompt Injection in Self-Healing LLM Agent Frameworks”

We are pleased to submit the above-referenced manuscript for consideration by *Empirical Software Engineering*.

Summary. This paper studies recovery-channel prompt injection: externally originated failure content is captured by a self-healing agent and re-injected into a subsequent LLM turn under a recovery frame that can be interpreted as authoritative. We combine systematic multiple-case source-code analysis with native-framework experiments. In the canonical Reflexion matrix, five model aliases produce attack success rates from 0.174 to 0.796 while all matched benign controls are zero. Native AutoGen and Aider executions provide additional framework evidence, and supplementary Reflexion contrast, defense-aware, and legitimate-recovery screens delimit what can and cannot be inferred at the completed sample sizes. A native Reflexion TrustOrigin port reduces attack success from 0.776 to 0.548 with zero matched benign false positives. The paper’s design implication is to pair provenance handling with structural authorization checks at recovery sinks.

We do not claim a new underlying language-model failure mode. The contribution is a software-engineering characterization of a distinct framework transport path and trust boundary: externally influenced failure content is captured, reframed, and re-injected by recovery infrastructure. The unit of analysis is the fixed-revision recovery implementation, and the results are reported for maintainers and designers of agent frameworks.

Contribution type. This is an empirical software-engineering contribution. The source analysis is reported as systematic multiple-case source-code analysis with qualitative coding, not as a complete Grounded Theory study. Quantitative results retain framework/model/payload strata, matched benign controls, Wilson confidence intervals, Fisher exact tests where appropriate, raw logs, and an explicit power analysis. Provider aliases and endpoint metadata are disclosed; the provider does not expose immutable checkpoint revisions, so exact snapshot reproducibility is stated as a limitation rather than implied.

Fit with EMSE. The manuscript addresses a software-engineering design assumption in recovery paths of real open-source agent frameworks and evaluates that assumption with reproducible source evidence and controlled execution traces. The replication package preserves canonical manifests, raw logs, endpoint verification, model provenance, and the numerical audit used to generate the tables. A fixed local Qwen2-family `qwen:7b` run is included as an open-model sensitivity anchor (100 valid trials); it is reported as exploratory evidence and does not replace the provider-alias matrix or imply an immutable snapshot for the hosted endpoints.

We confirm that this manuscript has not been published and is not under consideration elsewhere. We look forward to the editors’ and reviewers’ feedback.

Sincerely,

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